

Generation Interconnection System Impact Study Report

For Queue Project Q21-1934

Isabella 100 kV

May 30, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1934. The Interconnection Customer proposes a 92.3 MW battery energy storage facility interconnecting at the Isabella 100 kV substation in the Ohio area. The request was received on November 17, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 565 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$54,359,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mariposa - Mosher 100.0 kV	Barrett - Norwich 500.0 kV L	133.4	179	5.2
John C. Eisinger 100/1 kV Auto	Omni-Hackberry - Norwich 500.0 kV Line	132.1	316	28.6
Isabella - John C. Eisinger 100.0 kV	East Broad St - West Jct 100.0 kV	131.6	170	69.5
John C. Eisinger - E Cortez 100.0 kV line	Omni-Hackberry - Norwich 500.0 kV	129.2	126	14.1
Jug St - W Junction 100.0 kV	Isabella - John C. Eisinger 100.0 kV line	127.9	132	42.1
Mariposa - Acec Brooks 100.0 kV Line	Barrett - Norwich 500.0 kV	127.4	205	5.2
East Broad St - W Junction 100.0 kV	Isabella Substation - John C. Eisiner 100.0 kV	126.3	158	42.1
John C. Eisinger 230/1kV TX	Omni-Hackberry Substation - Norwich 500.0 kV	121.8	342	28.6
Penryn - John C. Eisinger 100.0 kV	Denmark - Thrive Fontana 100.0 kV line	108.2	133	8.6
Fleming Island - W Coshocton 100.0 kV	Omni-Hackberry - Norwich 500.0 kV	102.4	131	9.4
Opp - Penrn 100.0 kV line	Denmark - Thrive Fontana 100.0 kV	102.0	128	8.6
Vicco Substation - Mosher 100.0 kV L	Monterey Naval Lab - Neligh 100.0 kV line	100.4	151	7.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mariposa - Mosher 100.0 kV	Rebuild line with higher-capacity conductor	\$1,975,000	100% (\$1,975,000)
John C. Eisinger 100/1 kV Auto	Replace transformer with higher-rated unit	\$3,617,000	100% (\$3,617,000)
Isabella Substation - John C. Eisinger 100.0 kV	Rebuild line with higher-capacity conductor	\$2,971,000	100% (\$2,971,000)
John C. Eisinger - E Cortez 100.0 kV line	Rebuild line with higher-capacity conductor	\$15,163,000	100% (\$15,163,000)
Jug St - W Junction 100.0 kV	Rebuild line with higher-capacity conductor	\$4,417,000	100% (\$4,417,000)
Mariposa - Acec Brooks 100.0 kV Line	Rebuild line with higher-capacity conductor	\$2,468,000	100% (\$2,468,000)
W Jct - E Broad St 100.0 kV	Rebuild line with higher-capacity conductor	\$8,756,000	100% (\$8,756,000)
John C. Eisinger 230/1 kV Autotransformer	Replace transformer with higher-rated unit	\$9,657,000	100% (\$9,657,000)
Penryn - John C. Eisinger 100.0 kV	Reconductor line	\$1,145,000	100% (\$1,145,000)
Fleming Island - W Coshocton 100.0 kV	Reconductor line	\$2,581,000	40% (\$1,043,000)
Opp - Penryn 100.0 kV	Reconductor line	\$1,677,000	100% (\$1,677,000)
Vicco Substation - Mosher 100.0 kV L	Reconductor line	\$1,470,000	100% (\$1,470,000)
	Total allocated to project		\$54,359,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.